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Tang et al.

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(54) **TOUCH PANEL WITH WELL-DESIGNED TOUCH PATTERN**

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G06F 3/044 (2006.01)

(52) **U.S. Cl.**
CPC **G06F 3/0448** (2019.05)

(58) **Field of Classification Search**
None
See application file for complete search history.

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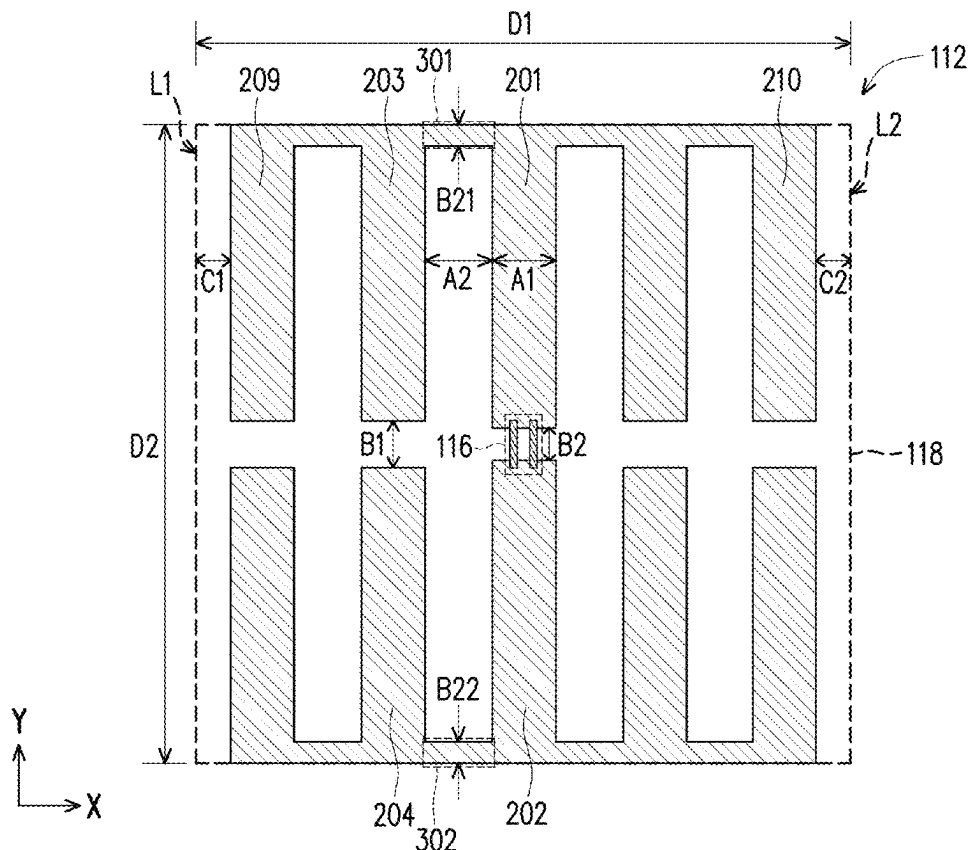
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(57) **ABSTRACT**

The touch panel includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of bridge structures. Each of the first electrodes includes a first branch, a second branch, a third branch and a fourth branch. The first and second branches are opposite to each other, and the third and fourth branches are opposite to each other. Each of the second electrodes includes a fifth branch, a sixth branch, a seventh branch and an eighth branch. The fifth and sixth branches are adjacently connected to each other, and the seventh and eighth branches are adjacently connected to each other. Each of the bridge structures connects the first and second branches of the first electrode. The third branch, the seventh branch, the first branch and the fifth branch are alternately arranged, and the fourth branch, the eighth branch, the second branch and the sixth branch are alternately arranged.

11 Claims, 11 Drawing Sheets



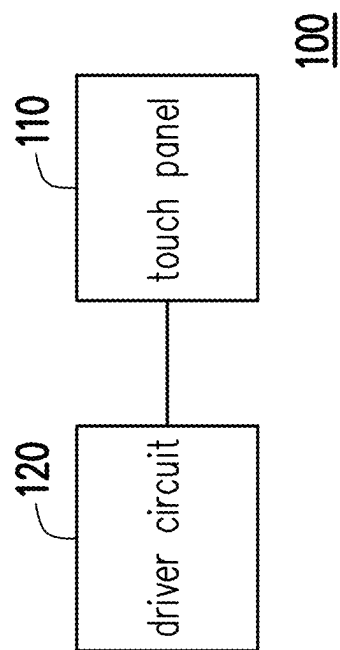


FIG. 1

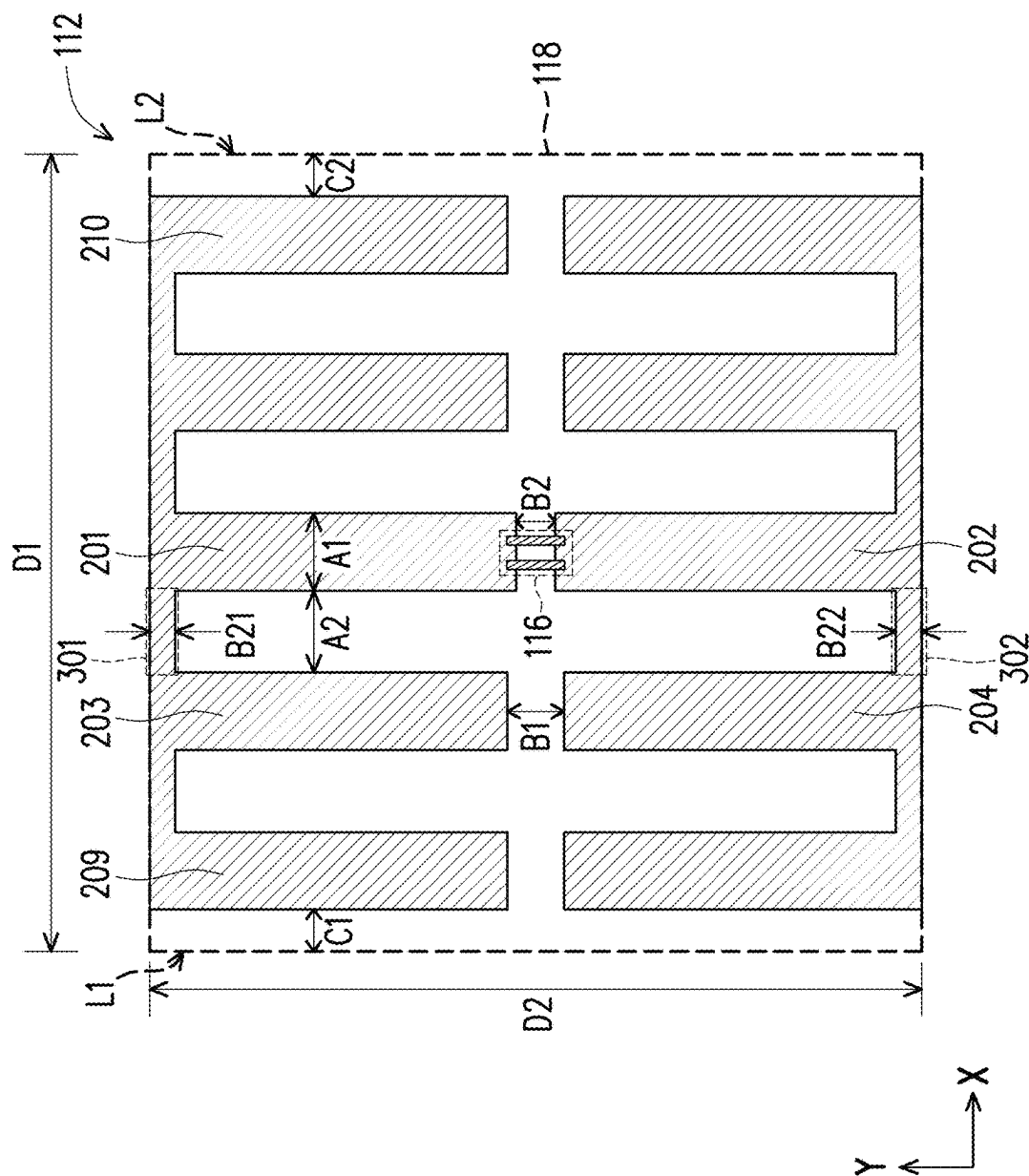


FIG. 2A

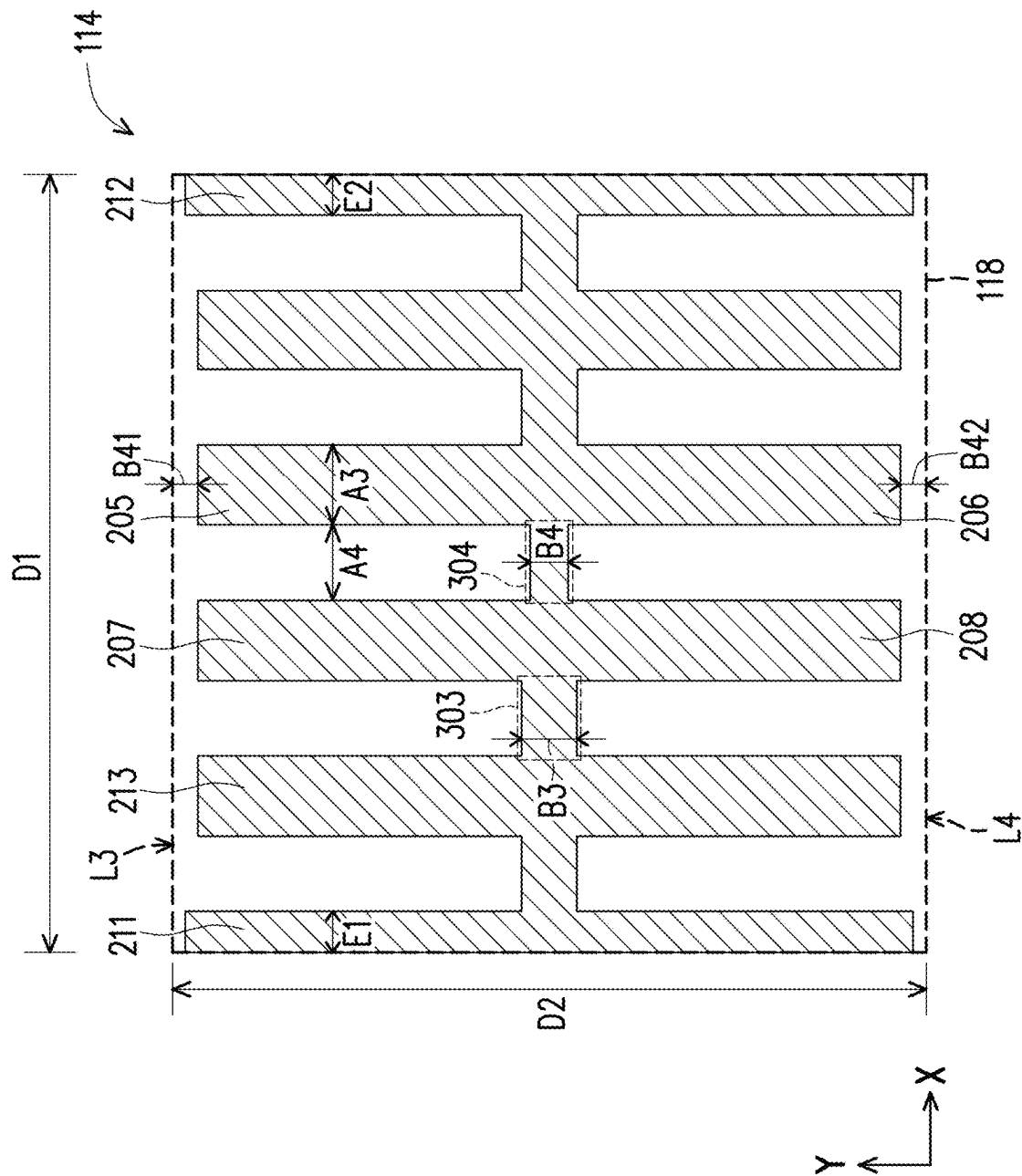


FIG. 2B

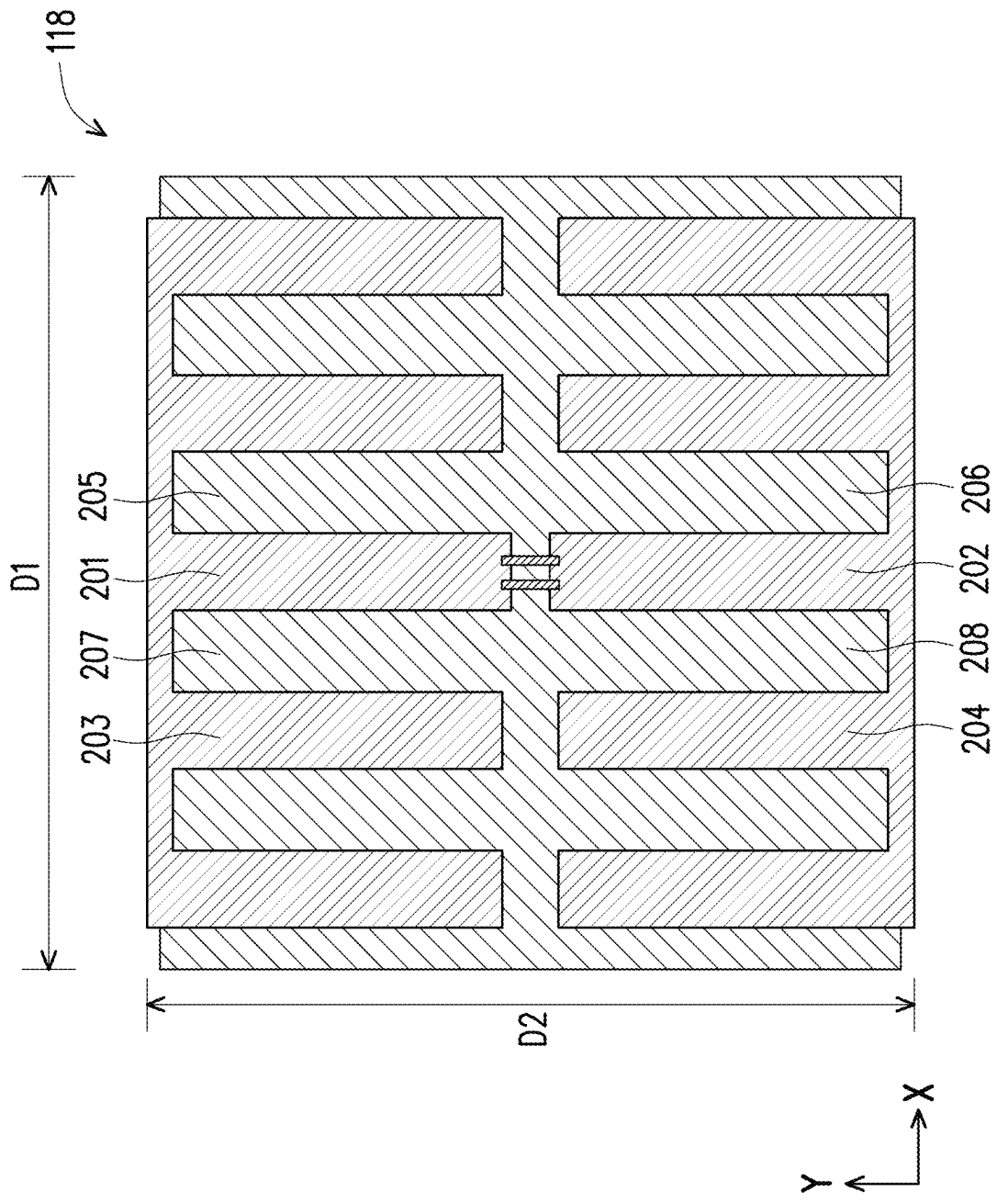


FIG. 3

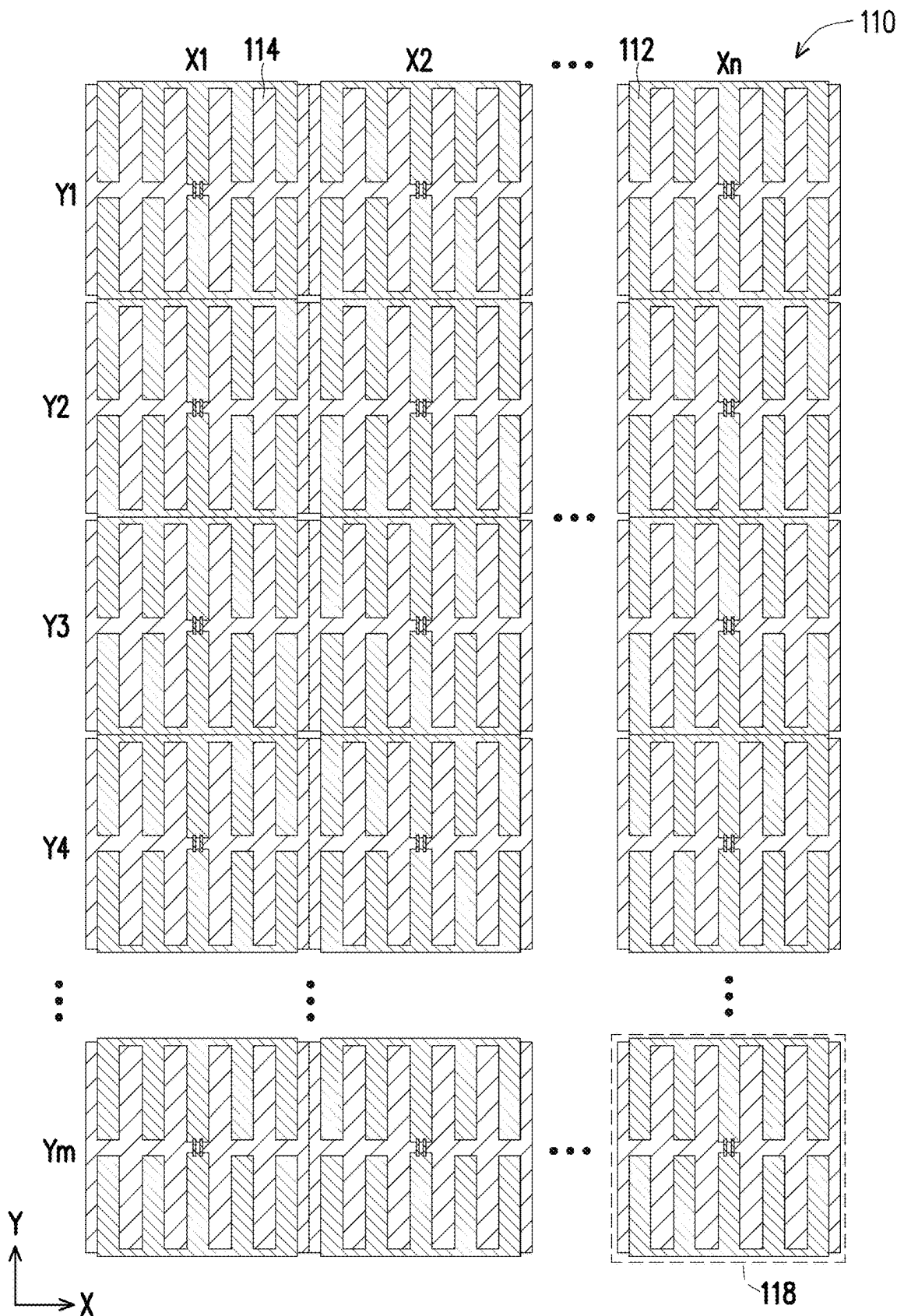


FIG. 4

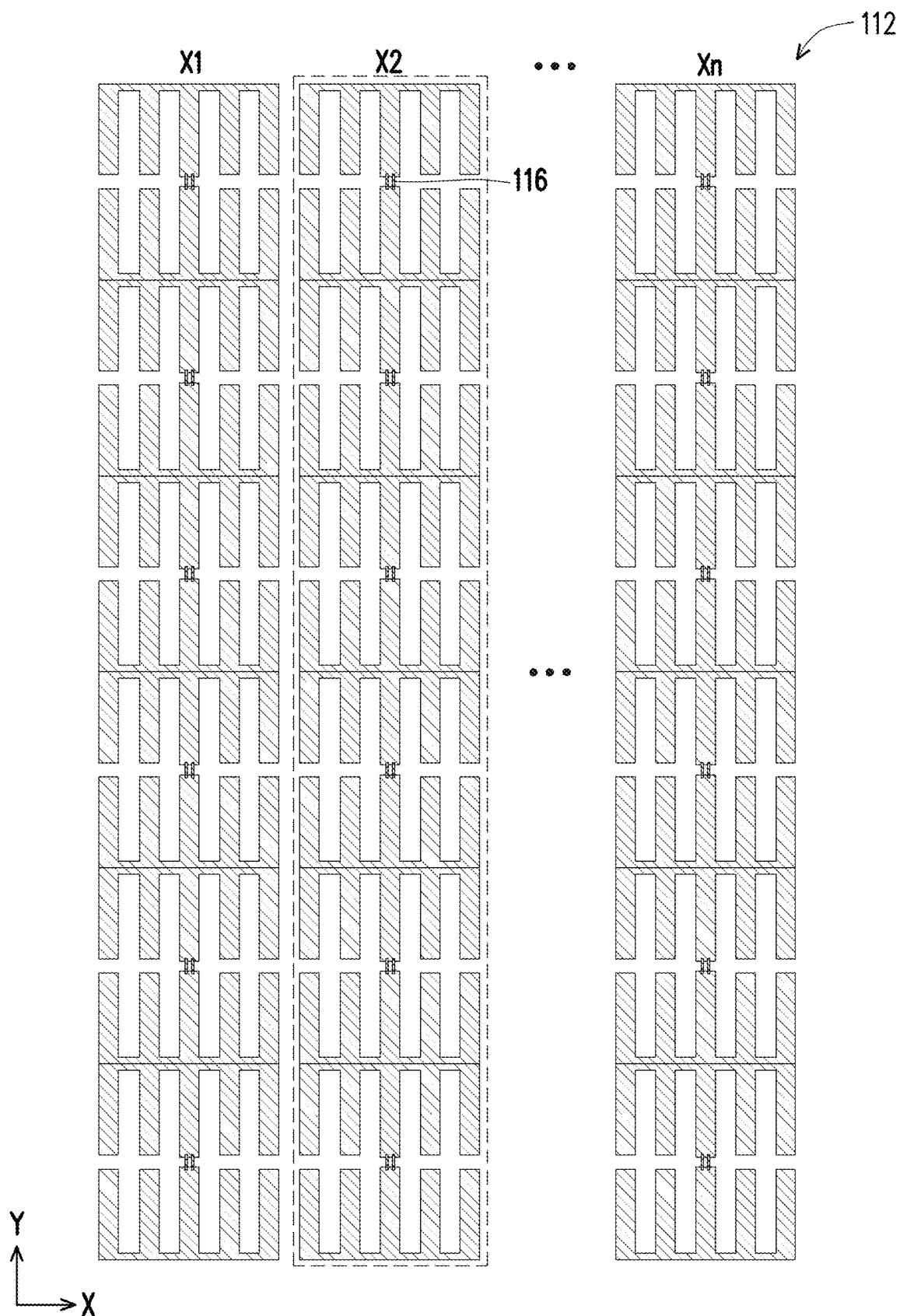


FIG. 5A

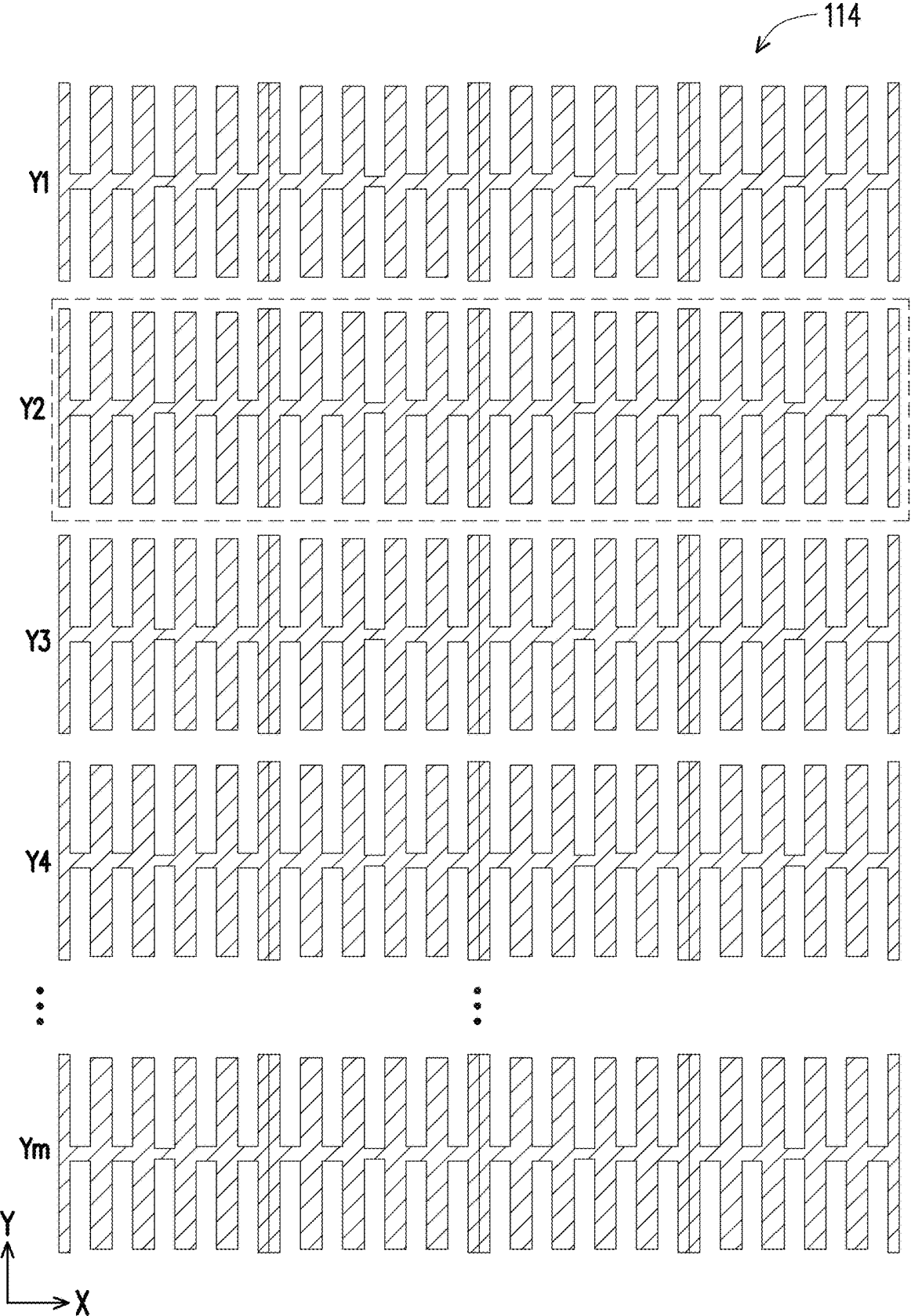


FIG. 5B

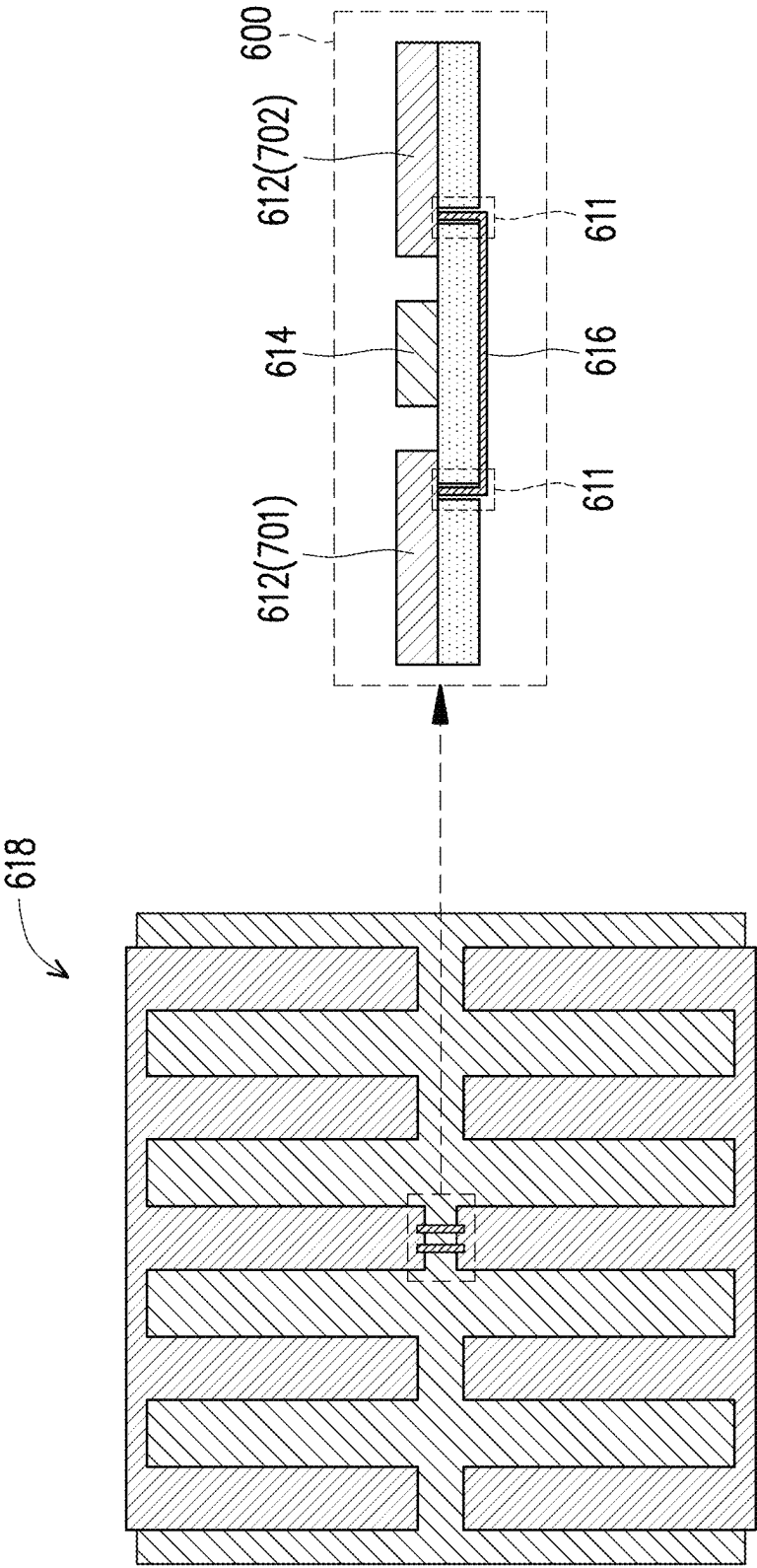


FIG. 6

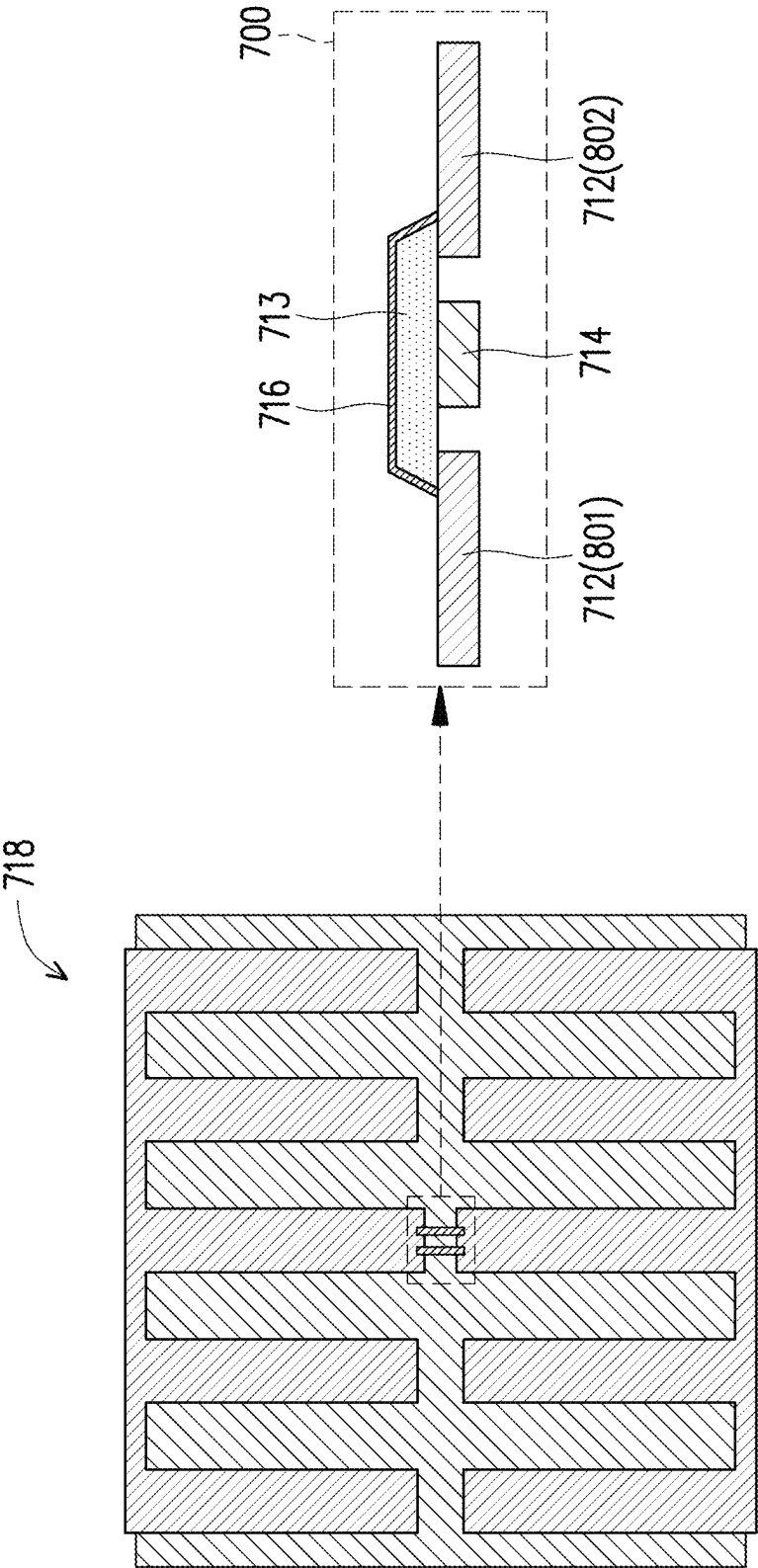


FIG. 7

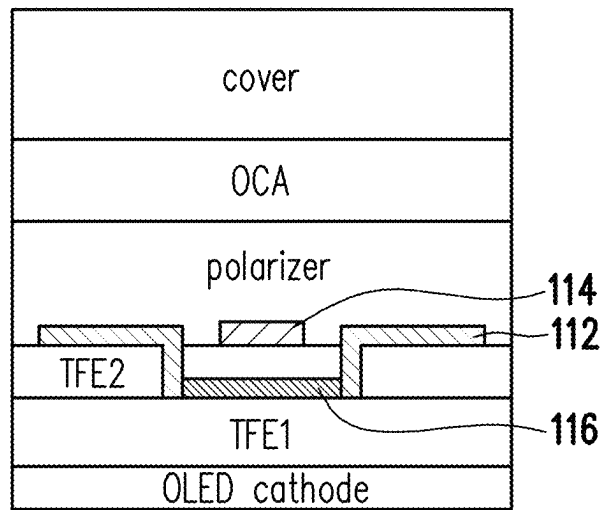


FIG. 8

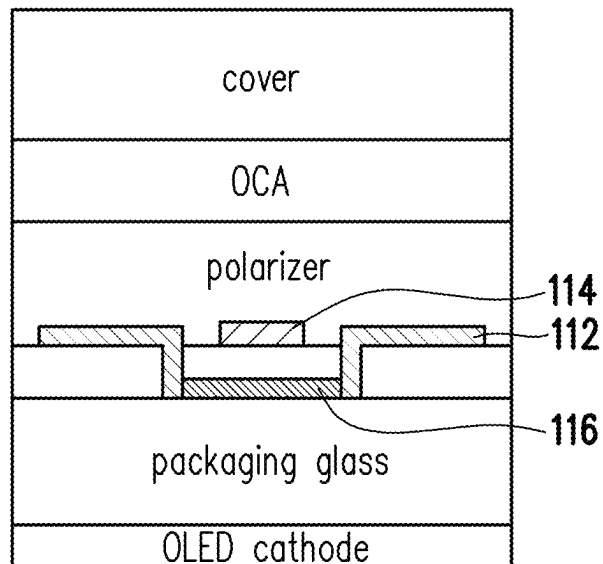


FIG. 9

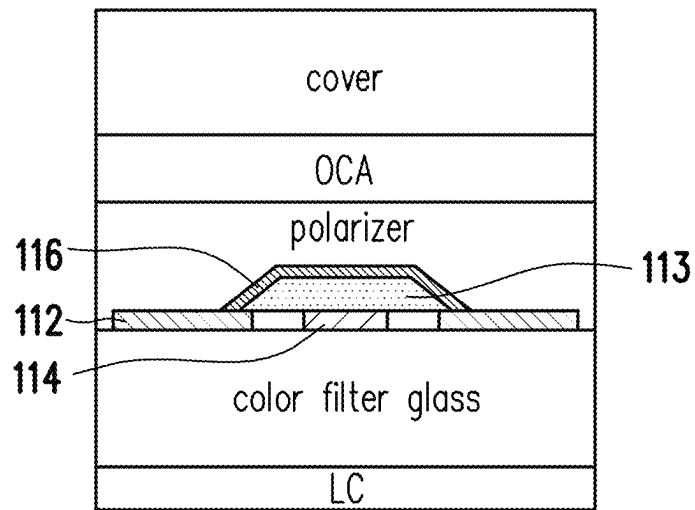


FIG. 10

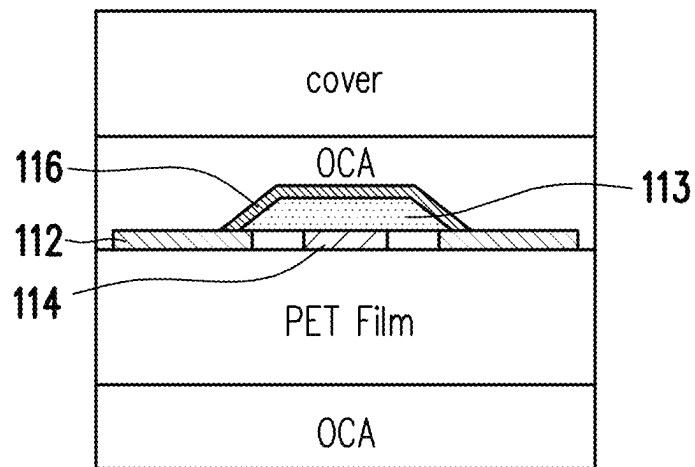


FIG. 11

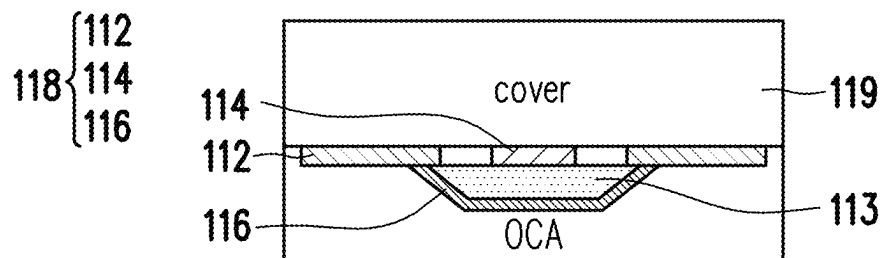


FIG. 12

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TOUCH PANEL WITH WELL-DESIGNED TOUCH PATTERN

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of U.S. provisional application Ser. No. 63/653,227, filed on May 30, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

The invention relates to a panel, and in particular, to a touch panel with a well-designed touch pattern.

Description of Related Art

Capacitive touch panels with mutual capacitance are capable of detecting changes in mutual capacitance. This is achieved by the mutual capacitance of the electrode pattern of the touch sensor when touched, thereby determining the behavior of the touch and the location of the touch. This detection technology is widely employed in input interfaces for mobile devices.

The design of the mutual capacitance touch pattern has a significant impact on its performance. A well-designed touch pattern can enhance the performance of a touch panel. Traditional touch patterns, such as the snowflake pattern and the diamond pattern, have been found to have limitations in terms of sensitivity and signal degradation in the case of poor grounding.

In addition, the traditional pattern doesn't work well in foldable covers because it doesn't have the versatility for different structures.

SUMMARY

The invention is directed to a touch panel, capable of increasing signal intensity and reducing signal degradation.

An embodiment of the invention providing a touch panel, including a plurality of first electrodes, a plurality of second electrodes, and a plurality of bridge structures. Each of the first electrodes includes a first branch, a second branch, a third branch and a fourth branch. The first branch and the second branch are opposite to each other, and the third branch and the fourth branch are opposite to each other. Each of the second electrodes includes a fifth branch, a sixth branch, a seventh branch and an eighth branch. The fifth branch and the sixth branch are adjacently connected to each other, and the seventh branch and the eighth branch are adjacently connected to each other. Each of the bridge structures connects the first branch and the second branch of the first electrode. The third branch, the seventh branch, the first branch and the fifth branch are alternately arranged, and the fourth branch, the eighth branch, the second branch and the sixth branch are alternately arranged.

To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated

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in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

FIG. 1 is a block diagram illustrating a touch apparatus according to an embodiment of the invention.

FIG. 2A and FIG. 2B are schematic diagrams illustrating a first electrode and a second electrode, respectively, according to an embodiment of the invention.

FIG. 3 is a schematic diagram illustrating a sensing unit according to an embodiment of the invention.

FIG. 4 illustrates a touch pattern arrangement on the touch panel according to an embodiment of the invention.

FIG. 5A and FIG. 5B respectively illustrate the first electrode arrangement and the second electrode arrangement of FIG. 4.

FIG. 6 illustrates a cross-section of a bridge area according to an embodiment of the invention.

FIG. 7 illustrates a cross-section of a bridge area according to another embodiment of the invention.

FIG. 8 shows a flexible organic light emitting diode (OLED) display with on-cell touch sensors according to another embodiment of the invention.

FIG. 9 shows an inflexible OLED display with on-cell touch sensors according to another embodiment of the invention.

FIG. 10 shows a liquid crystal display (LCD) with on-cell touch sensors according to another embodiment of the invention.

FIG. 11 shows an LCD with add-on touch sensors according to another embodiment of the invention.

FIG. 12 shows a structure of OGS (On Glass Solution) sensors according to another embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

The term "coupled (or connected)" used in this specification (including claims) may refer to any direct or indirect connection means. For example, "a first device is coupled (connected) to a second device" should be interpreted as "the first device is directly connected to the second device" or "the first device is indirectly connected to the second device through other devices or connection means". The terms "first" and "second" mentioned in the full text of the specification of the disclosure (including claims) are used to name elements or to distinguish different embodiments or scopes, neither to be used to limit upper or lower limit of the number of elements nor limit the sequence of the elements. In addition, wherever possible, elements/components/steps with the same reference numbers are used in the drawings and embodiments to represent the same or similar parts. Elements/components/steps using the same numbers or using the same terms in different embodiments may serve as cross-reference for each other.

FIG. 1 is a block diagram illustrating a touch apparatus according to an embodiment of the invention. Referring to FIG. 1, the touch apparatus 100 includes a touch panel 110 and a driver circuit 120. The driver circuit 120 is configured to drive the touch panel 110 to perform a touch sensing operation.

FIG. 2A and FIG. 2B are schematic diagrams illustrating a first electrode and a second electrode, respectively, according to an embodiment of the invention. FIG. 3 is a schematic diagram illustrating a sensing unit according to an embodiment of the invention.

Referring to FIG. 2A to FIG. 3, the touch panel 110 includes a plurality of first electrodes 112, a plurality of

second electrodes **114** and a plurality of bridge structures **116**. The first electrode **112** and the second electrode **114** has complementary shapes and can be combined to form a sensing unit **118**. The touch panel **110** includes a plurality of sensing pads to perform the touch sensing operation. The sensing unit **118** may serve as a sensing pad of the touch panel **110**.

As shown in FIG. 2A and FIG. 2B, the first electrode **112** and the second electrode **114** are different fence-like electrodes. The first electrode **112** at least includes a first branch **201**, a second branch **202**, a third branch **203** and a fourth branch **204**. The first branch **201**, the second branch **202**, the third branch **203** and the fourth branch **204** are arranged in a first direction X and extended in a second direction Y. The first branch **201** and the second branch **202** are opposite to each other in the second direction Y, and the third branch **203** and the fourth branch **204** are opposite to each other on another line in the second direction Y.

The bridge structure **116** connects the first branch **201** and the second branch **202** to make two portions of the first electrode **112** to be conductive. In the present embodiment, two conductive strip electrodes are used as an example of the bridge structure **116**, but the shape and number of the electrodes are not intended to limit the invention.

The second electrode **114** at least includes a fifth branch **205**, a sixth branch **206**, a seventh branch **207** and an eighth branch **208**. The fifth branch **205**, the sixth branch **206**, the seventh branch **207** and the eighth branch **208** are arranged in the first direction X and extended in the second direction Y. The fifth branch **205** and the sixth branch **206** are adjacently connected to each other in the second direction Y, and the seventh branch **207** and the eighth branch **208** are adjacently connected to each other on another line in the second direction Y.

In the present embodiment, the first electrodes and the second electrodes are metal plate electrodes, flexible metal-mesh electrodes, or indium tin oxide (ITO) electrodes, but the invention is not limited thereto. In the present embodiment, the materials of the first electrodes and the second electrodes may be molybdenum (Mo), aluminum (Al), titanium (Ti), copper (Cu), silver (Ag), or an alloy thereof, but the invention is not limited thereto.

As shown in FIG. 3, the first electrode **112** and the second electrode **114** are combined to form the sensing unit **118**. The sensing unit **118** has a first pitch **D1** in the first direction X and a second pitch **D2** in the Y-direction. The first pitch **D1** and the second pitch **D2** comply with a condition of $0.8 < D1/D2 < 1.2$. In the sensing unit **118**, the third branch **203**, the seventh branch **207**, the first branch **201** and the fifth branch **205** are alternately arranged from left to right in the first direction X. Similarly, the fourth branch **204**, the eighth branch **208**, the second branch **202** and the sixth branch **206** are also alternately arranged from left to right in the first direction X.

The structural parameters of the first electrode **112** and the second electrode **114** are described below.

Referring to FIG. 2A, a width of the first branch **201** in the first direction X is **A1**, and an interval **A2** between the first branch **201** and the third branch **203** in the first direction X is **A2**. The width **A1** and the interval **A2** comply with a condition of $0.5 < A1/A2 < 2$. An interval between the third branch **203** and the fourth branch **204** in the second direction Y is **B1**, and an interval between the first branch **201** and the second branch **202** in the second direction Y is **B2**, and the intervals **B1** and **B2** comply with a condition of $0.5 < B1/B2 < 2$.

The first electrode **112** further includes a first segment **301** and a second segment **302** arranged in the second direction Y and extended in the first direction X. The first segment **301** is located between the first branch **201** and the third branch **203**. The second segment **302** is located between the second branch **202** and the fourth branch **204**. A width of the first segment **301** in the second direction Y is **B21**, and a width of the second segment **302** in the second direction Y is **B22**. The interval **B1** and the widths **B21** and **B22** comply with a condition of $B1 = B21 = B22$.

In addition, the sensing unit **118** has a first edge line **L1** and a second edge line **L2** arranged in the first direction X, and the first electrode **112** further includes a ninth branch **209** and a tenth branch **210** arranged in the first direction X. An interval between the ninth branch **209** and the first edge line **L1** in the first direction X is **C1**, and an interval between the tenth branch **210** and the second edge line **L2** in the first direction X is **C2**. The intervals **A2**, **C1** and **C2** comply with a condition of $C1 + C2 = A2$.

Referring to FIG. 2B, a width of the fifth branch **205** in the first direction X is **A3**, and an interval between the fifth branch **205** and the seventh branch **207** in the first direction X is **A4**. The width **A3** and the interval **A4** comply with a condition of $0.5 < A3/A4 < 2$. The second electrode **114** further includes an eleventh branch **211**, a twelfth branch **212** and a thirteenth branch **213** arranged in the first direction X. The second electrode **114** further includes a third segment **303** located between the seventh branch **207** and the thirteenth branch **213** and a fourth segment **304** located between the fifth branch **205** and the seventh branch **207**. A width of the third segment **303** in the second direction Y is **B3**, and a width of the fourth segment **304** in the second direction Y is **B4**, and the widths **B3** and **B4** comply with a condition of $0.5 < B3/B4 < 2$.

The sensing unit **118** has a third edge line **L3** and a fourth edge line **L4** arranged in the second direction Y. An interval between the fifth branch **205** and the third edge line **L3** in the second direction Y is **B41**, and an interval between the sixth branch **206** and the fourth edge line **L4** in the second direction Y is **B42**. The width **B4** and the intervals **B41** and **B42** comply with a condition of $B4 = B41 = B42$.

In addition, a width of the eleventh branch **211** in the first direction X is **E1**, and a width of the twelfth branch **212** in the first direction is **E2**. The widths **A3**, **E1** and **E2** comply with a condition of $E1 + E2 = A3$.

In a case where the shapes of the first electrode **112** and the second electrode **114** are complementary, the structural parameters **A1**, **A2**, **A3** and **A4** comply with the conditions of $A2 \geq A3$ and $A4 \geq A1$.

In the present embodiment, the first electrode **112** is configured to have ten identical branches and the second electrode **114** is configured to have eight identical branches. However, the number and shape of the branches of each electrode are not intended to limit the invention.

In order to optimize the touch sensing intensity, the width and the ratio of the branches of the first electrodes and the second electrodes can be designed according to the above structural parameters. However, the above structural parameters of the first electrode **112** and the second electrode **114** are by way of example, but are not intended to limit the invention.

FIG. 4 illustrates a touch pattern arrangement on the touch panel according to an embodiment of the invention. FIG. 5A and FIG. 5B respectively illustrates the first electrode arrangement and the second electrode arrangement of FIG. 4. Referring to FIG. 4 to FIG. 5B, the first electrodes **112** are connected to each other in the second direction Y through a

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plurality of bridge structures **116**, and form separate first channels **X1** to **Xn** in the first direction **X**. The second electrodes **114** are connected to each other in the first direction **X** through a plurality of segments **303**, **304**, and form separate second channels **Y1** to **Ym** in the second direction **Y**. The first electrodes **112** and the second electrodes **114** form $n \times m$ mutual capacitance nodes, where n and m are positive integers.

In the present embodiment, the first electrodes **112** serve as driving electrodes, and the second electrodes **114** serve as sensing electrodes. The driver circuit **120** sequentially drives the first channels from **X1** to **Xn**, and receives sensing signals from the second channels **Y1** to **Ym**.

FIG. **6** illustrates a cross-section of a bridge area **600** according to an embodiment of the invention. Referring to FIG. **6**, each sensing unit **618** of the touch pattern includes a first electrode **612** and a second electrode **614**. Two branches **701** and **702** of the first electrode **612** are connected by a bridge structure **616**. The first electrode **612** and the second electrode **614** are on the same conductor layer, and the bridge structure **616** is on a different conductor layer. In the present embodiment, the bridge structure **616** crosses the second electrode **614** by means of through holes **611** and connects the separate branches **701** and **702**.

FIG. **7** illustrates a cross-section of a bridge area **700** according to another embodiment of the invention. Referring to FIG. **7**, for a sensing unit **718**, the bridge structure **716** crosses the second electrode **714** by means of an insulating island **713** and connects the separate branches **801** and **802** of the first electrode **712**.

The touch pattern of the disclosure may be applied to the following structures, but they are not intended to limit the invention.

FIG. **8** shows a flexible organic light emitting diode (OLED) display with on-cell touch sensors according to another embodiment of the invention. FIG. **9** shows an inflexible OLED display with on-cell touch sensors according to another embodiment of the invention. In FIG. **8** and FIG. **9**, the devices TFE1 and TFE2 are thin film encapsulation layers, and the device OCA is optically clear adhesives. Taking FIG. **2A** and FIG. **2B** as an example, the first electrode **112**, the second electrode **114** and the bridge structure **116** can be applied to the display structures of FIG. **8** and FIG. **9**. The bridge structure **116** crosses the second electrode **114** by means of through holes and connects the separate branches of the first electrode **112**.

FIG. **10** shows a liquid crystal display (LCD) with on-cell touch sensors according to another embodiment of the invention. FIG. **11** shows an LCD with add-on touch sensors according to another embodiment of the invention. In FIG. **10** and FIG. **11**, the bridge structure **116** crosses the second electrode **114** by means of an insulating island **113** and connects to the separate branches of the first electrode **112**.

FIG. **12** shows a structure of OGS (On Glass Solution) sensors according to another embodiment of the invention. In FIG. **12**, the sensing unit **118** serves as a touch sensor and is disposed on a cover glass **119**.

In summary, in the embodiments of the invention, the touch patterns have higher mutual capacitance change and lower signal attenuation ratio in the case of poor grounding in both straight and foldable architectures. This makes it possible for flexible panels with the proposed touch patterns to achieve cost reduction without having to change the touch patterns due to changes in the cover structures of the end products.

It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed

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embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

What is claimed is:

1. A touch panel, comprising:

a plurality of first electrodes, each comprising a first branch, a second branch, a third branch and a fourth branch, wherein the first branch and the second branch are opposite to each other, and the third branch and the fourth branch are opposite to each other;

a plurality of second electrodes, each comprising a fifth branch, a sixth branch, a seventh branch and an eighth branch, wherein the fifth branch and the sixth branch are adjacently connected to each other, and the seventh branch and the eighth branch are adjacently connected to each other; and

a plurality of bridge structures, each connecting the first branch and the second branch of the first electrode, wherein the third branch, the seventh branch, the first branch and the fifth branch are alternately arranged, and the fourth branch, the eighth branch, the second branch and the sixth branch are alternately arranged,

wherein the first electrodes and the second electrodes form a plurality of sensing units, and each of the sensing units comprises one first electrode and one second electrode,

wherein each of the sensing unit has a first pitch **D1** in a first direction and a second pitch **D2** in a second direction, and the first pitch **D1** and the second pitch **D2** comply with a condition of $0.8 < D1/D2 < 1.2$,

wherein a width of the first branch in the first direction is **A1**, an interval between the first branch and the third branch in the first direction is **A2**, and the width **A1** and the interval **A2** comply with a condition of $0.5 < A1/A2 < 2$.

2. The touch panel according to claim 1, wherein the first branch, the second branch, the third branch and the fourth branch are arranged in the first direction and extended in the second direction.

3. The touch panel according to claim 1, wherein the fifth branch, the sixth branch, the seventh branch and the eighth branch are arranged in the first direction and extended in the second direction.

4. The touch panel according to claim 1,

wherein the sensing unit has a first edge line and a second edge line arranged in the first direction, and each of the first electrodes further comprises a ninth branch and a tenth branch arranged in the first direction,

wherein an interval between the ninth branch and the first edge line in the first direction is **C1**, an interval between the tenth branch and the second edge line in the first direction is **C2**, and the intervals **A2**, **C1** and **C2** comply with a condition of $C1 + C2 = A2$.

5. The touch panel according to claim 1, wherein an interval between the third branch and the fourth branch in the second direction is **B1**, an interval between the first branch and the second branch in the second direction is **B2**, and the intervals **B1** and **B2** comply with a condition of $0.5 < B1/B2 < 2$.

6. The touch panel according to claim 5,

wherein each of the first electrodes further comprises a first segment and a second segment arranged in the second direction and extended in the first direction, the first segment is located between the first branch and the

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third branch, and the second segment is located between the second branch and the fourth branch,

wherein a width of the first segment in the second direction is **B21**, a width of the second segment in the second direction is **B22**, and the interval **B2** and the widths **B21** and **B22** comply with a condition of $B2=B21=B22$.

7. The touch panel according to claim 1, wherein a width of the fifth branch in the first direction is **A3**, an interval between the fifth branch and the seventh branch in the first direction is **A4**, and the width **A3** and the interval **A4** comply with a condition of $0.5<A3/A4<2$.

8. The touch panel according to claim 7,

wherein each of the second electrodes further comprises an eleventh branch and a twelfth branch arranged in the first direction,

wherein a width of the eleventh branch in the first direction is **E1**, a width of the twelfth branch in the first direction is **E2**, and the widths **A3**, **E1** and **E2** comply with a condition of $E1+E2=A3$.

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9. The touch panel according to claim 7,

wherein each of the second electrodes further comprises a thirteenth branch arranged in the first direction, a third segment located between the seventh branch and the thirteenth branch, and a fourth segment located between the fifth branch and the seventh branch,

wherein a width of the third segment in the second direction is **B3**, a width of the fourth segment in the second direction is **B4**, and the widths **B3** and **B4** comply with a condition of $0.5<B3/B4<2$.

10. The touch panel according to claim 9,

wherein the sensing unit has a third edge line and a fourth edge line arranged in the second direction,

wherein an interval between the fifth branch and the third edge line in the second direction is **B41**, an interval between the sixth branch and the fourth edge line in the second direction is **B42**, and the width **B4** and the intervals **B41** and **B42** comply with a condition of $B4=B41=B42$.

11. The touch panel according to claim 1, wherein the first electrodes and the second electrodes are different fence-like electrodes.

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